LAB 2 FINAL REPORT

Agile Backlog Creation & Sprint Simulation in Jira

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Program	B.Tech — Computer Science & Engineering
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Project	Municipal Infrastructure Damage Reporting Application
Jira Project Key	MN

This report documents the Jira Agile workflow completed for Lab 2, including backlog creation, Epics, User Stories, prioritization, Fibonacci story-point estimation, two sprint simulations, Burndown analysis, and reflection.

1. Objective

The objective was to use Jira to turn Lab 1 functional requirements into Agile backlog items, prioritize and estimate them, simulate sprint execution, and analyze progress using a Burndown Chart.

2. Project and Agile Backlog

The Municipal Infrastructure Damage Reporting Application is a smart-city citizen portal for reporting infrastructure problems such as potholes, broken streetlights, and water-pipeline leaks using geotagged photos and automatic ticket dispatch.

Epic	Name	Purpose
MN-1	Citizen Damage Reporting	Capture and submit infrastructure damage reports.
MN-2	Automatic Location & Ticket Dispatch	Identify incident location and route reports to the responsible department.
MN-3	Ticket Tracking & Resolution	Manage, track, and notify users about ticket progress.

ID	User Story	Epic	Priority	SP
MN-4	Submit Infrastructure Damage Report	MN-1	High	5
MN-5	Upload Damage Photo	MN-1	High	3
MN-6	Draft Report Offline	MN-1	High	5
MN-7	Extract GPS from Photo	MN-2	High	5
MN-8	Pinpoint Incident on GIS Map	MN-2	High	3
MN-9	Automatically Dispatch Ticket	MN-2	High	5
MN-10	Review and Update Ticket	MN-3	High	5
MN-11	Track Ticket Status	MN-3	Medium	3
MN-12	Receive Status Notifications	MN-3	Medium	3
MN-13	Reliable Fast Ticket Access	MN-3	High	5

Total backlog estimate: 42 story points. Estimates use Fibonacci-style relative sizing based on effort, complexity and uncertainty.

Backlog Evidence

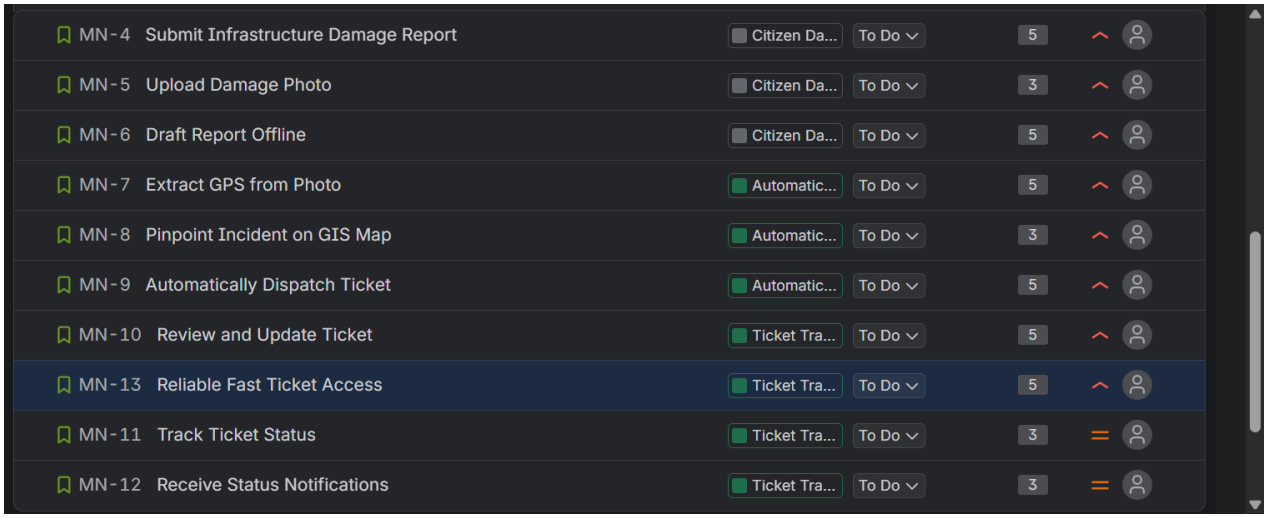


Figure 1. Complete Jira backlog showing all 10 User Stories, priorities, and Fibonacci story points.

3. User Story Details

MN-4

As a citizen reporter, I want to submit an infrastructure damage report so that the municipality can record and act on the reported issue.

MN-5

As a citizen reporter, I want to upload a photo of the infrastructure damage so that the issue can be visually verified.

MN-6

As a citizen reporter, I want to draft a damage report while offline so that I can save my report and automatically sync it when network connectivity resumes.

MN-7

As a citizen reporter, I want the system to extract GPS coordinates from the uploaded photo so that the incident location can be automatically identified.

MN-8

As a municipal engineer, I want the reported damage location to be pinpointed on the GIS map so that I can accurately identify where the infrastructure problem occurred.

MN-9

As a municipal engineer, I want damage reports to be automatically dispatched to the responsible engineering department so that each issue reaches the appropriate team for action.

MN-10

As a municipal engineer, I want to review and update assigned damage tickets so that I can manage the issue and record its progress toward resolution.

MN-11

As a citizen reporter, I want to track the status of my damage report so that I know whether the reported issue is being reviewed, assigned, or resolved.

MN-12

As a citizen reporter, I want to receive notifications when my ticket status changes so that I stay informed about the progress of my reported issue.

MN-13

As a citizen reporter, I want ticket information to load quickly and reliably so that I can access the status of my reported issue without unnecessary delays.

4. Sprint Planning and Execution

The handout requires Sprint 1 to be one week, work to progress To Do → In Progress → Done, and explicitly requires the completion of two sprints. The Jira project was therefore executed across Sprint 1 and Sprint 2.

Sprint	Stories	Points	Result
Sprint 1	MN-4, MN-5, MN-7, MN-9, MN-10, MN-11, MN-12	29	7 completed
Sprint 2	MN-6, MN-8, MN-13	13	3 completed
Total	10 stories	42	2 sprints completed

Sprint 1: Seven prioritized stories totaling 29 SP were simulated through To Do, In Progress and Done, and the sprint was completed.

Sprint 2: MN-6 (5 SP), MN-8 (3 SP), and MN-13 (5 SP) totaled 13 SP. All three were moved through To Do → In Progress → Done and Sprint 2 was completed.

5. Sprint 1 Evidence

The following Jira screenshot documents the Sprint 1 board after the seven selected stories were completed.

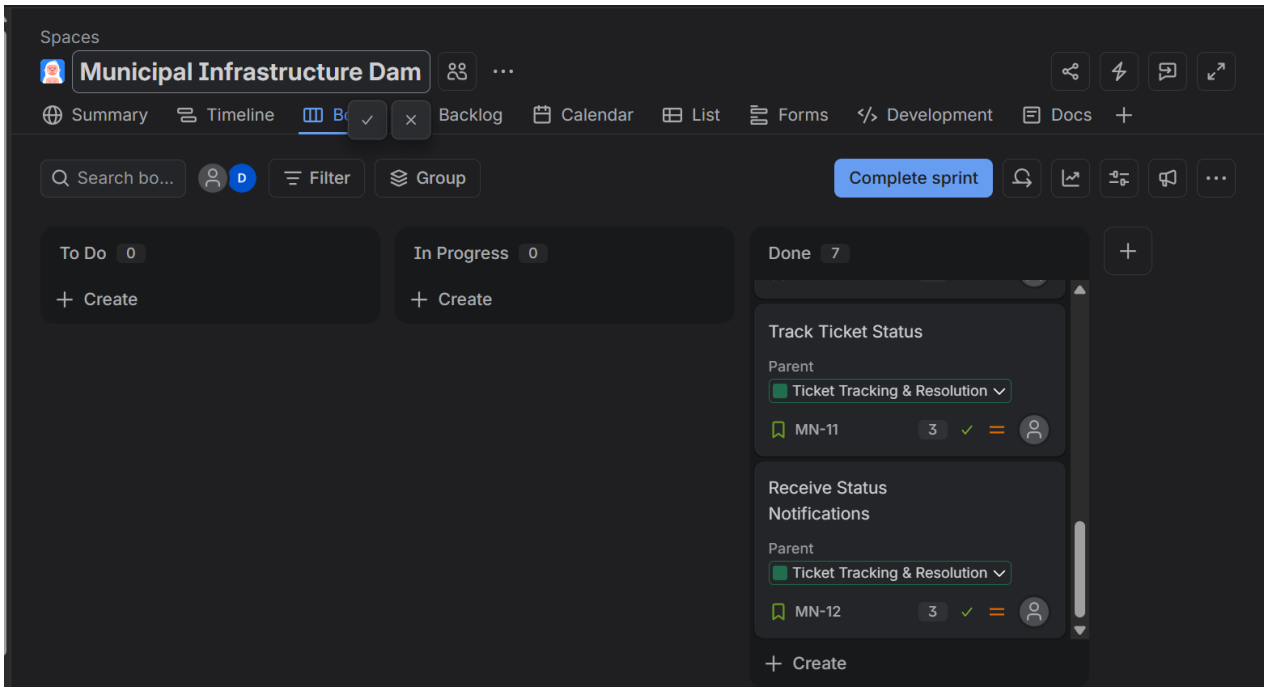


Figure 2. Sprint 1 Active Sprint board showing all 7 Sprint 1 stories in Done.

6. Sprint Burndown Evidence

The Burndown Charts below provide visual evidence for the sprint reporting requirement. Sprint 1 shows the September 2–8 reporting window, while Sprint 2 shows the September 9–16 reporting window and a starting scope of 13 story points.

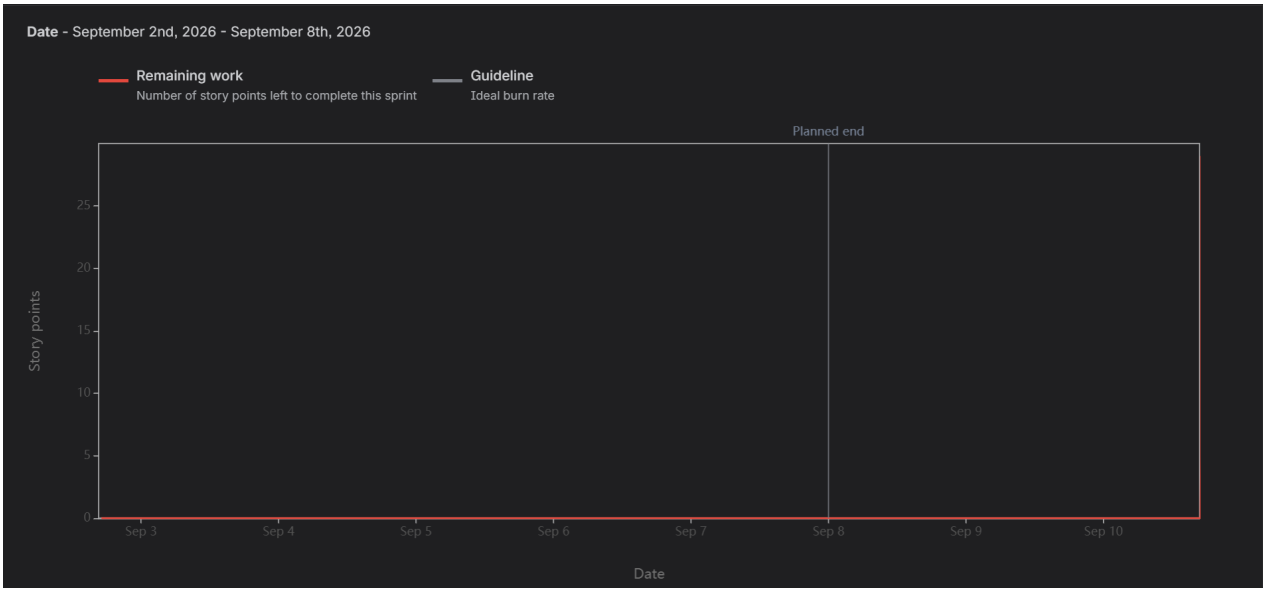


Figure 3. Sprint 1 Burndown Chart / remaining-work view for the September 2–8 sprint period.

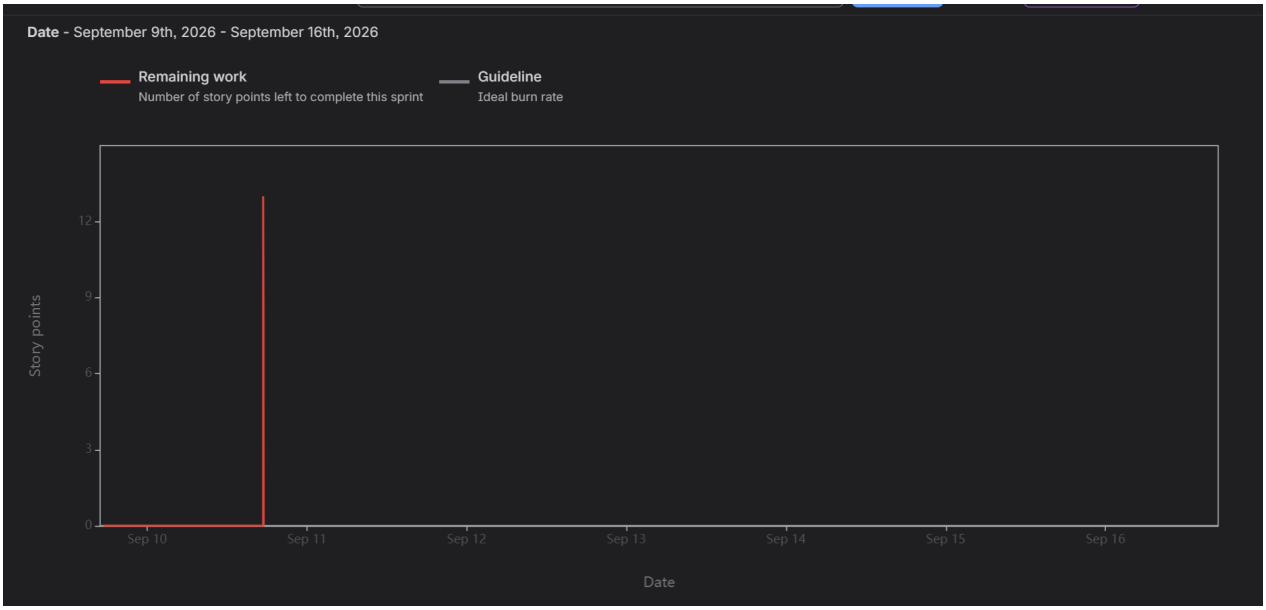


Figure 4. Sprint 2 Burndown Chart / remaining-work view for the September 9–16 sprint period.

7. Reflection

1. Did your estimations reflect the actual effort?

The Fibonacci estimates reasonably represented relative effort. Three-point stories were more focused, while five-point stories involved greater implementation effort or uncertainty. Because this was a simulated exercise, the estimates are best treated as relative estimates rather than real development-hour measurements.

2. Was your backlog well-prioritized?

Yes. The backlog was ordered mainly by user impact and functional importance. High-priority reporting, location, dispatch, ticket-management and reliability capabilities were placed ahead of the medium-priority tracking and notification stories.

3. How did your simulated sprint align with your plan?

The simulation followed the planned Scrum workflow. Sprint 1 contained seven stories totaling 29 SP and all seven were completed. Sprint 2 contained the remaining three stories totaling 13 SP and all three were completed. In both sprints, stories progressed through To Do → In Progress → Done.

4. What insights did the burndown chart give about your team's capacity?

The Sprint 2 Burndown Chart started with 13 story points of planned work and reached zero remaining work after the three stories were completed. This indicates that the simulated team completed its committed scope. The chart also shows how remaining work can be compared with the ideal guideline to understand progress. Since this was a simulation, it should not be treated as a real historical velocity measurement.

8. Conclusion

This laboratory demonstrated the Agile backlog and sprint-simulation workflow in Jira. The project was organized into three Epics and ten User Stories; priorities and Fibonacci story points were assigned; two sprints were completed; and the Burndown Chart was used to analyze remaining work and sprint progress.

9. Submission Checklist

✓ Jira backlog with Epics and User Stories

✓ Priority assignments

✓ Fibonacci story-point assignments

✓ Sprint 1 Active Sprint evidence

✓ Sprint 2 execution and completion

✓ Completion of 2 sprints

✓ Sprint 1 and Sprint 2 Burndown Chart evidence

✓ Reflection answers

✓ Jira workspace ready for instructor demonstration